Rational Functions: Practice Test

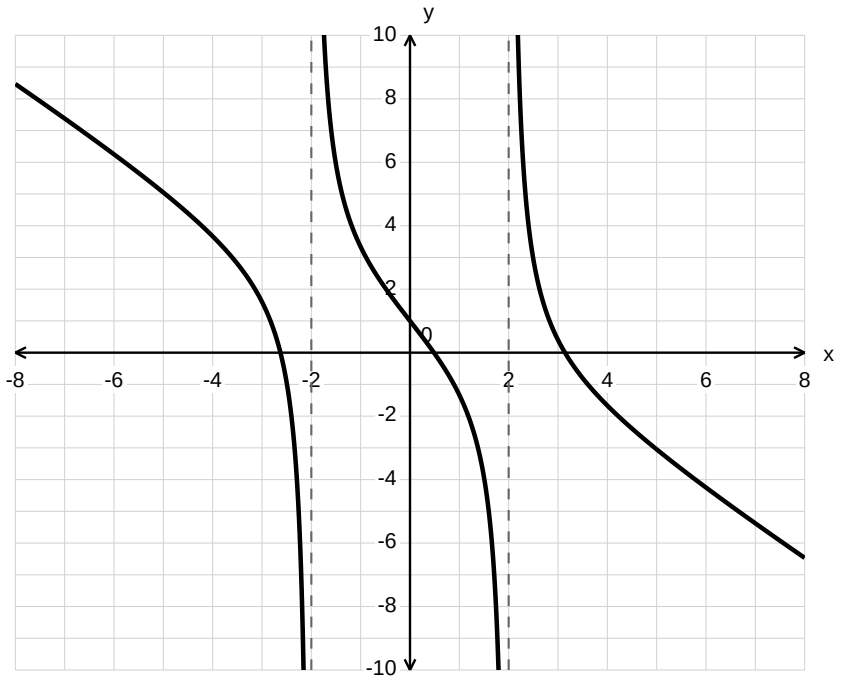
Name: _____ Date: _____

5 problems | About 25 minutes | Four-function calculator allowed

Show your work and explain your choices. Use the answer space provided.

Problem 1. Read and explain the graph

The graph of a rational function *h* is shown below.



(a) Domain (interval notation): _____

(b) Explain what happens near each excluded *x*-value and as you move far to the left and far to the right on the graph.

Your explanation

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Problem 2. Calculate a slant asymptote

Use polynomial long division to find the slant asymptote.
Show all your work.

$$f(x) = \frac{x^3 + 3x^2 - 2x + 7}{x^2 - 4x + 2}$$

Long division

Slant asymptote: _____

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Problem 3. Apply a concentration model

A filter removes a dissolved chemical from a tank of water.
The chemical's concentration in the water is modeled by

$$C(t) = \frac{420}{t^2 + 4t + 24}$$

The model applies for the first 12 minutes. Here t is minutes since the filter started, and $C(t)$ is concentration in milligrams per liter (mg/L).

(a) Explain why *C* is a rational function.

Your explanation

(b) Calculate *C*(4). Show the substitution and arithmetic.

Your calculation

(c) Write a complete-sentence interpretation of *C*(4). Explain what the input and output mean in this situation. Include units.

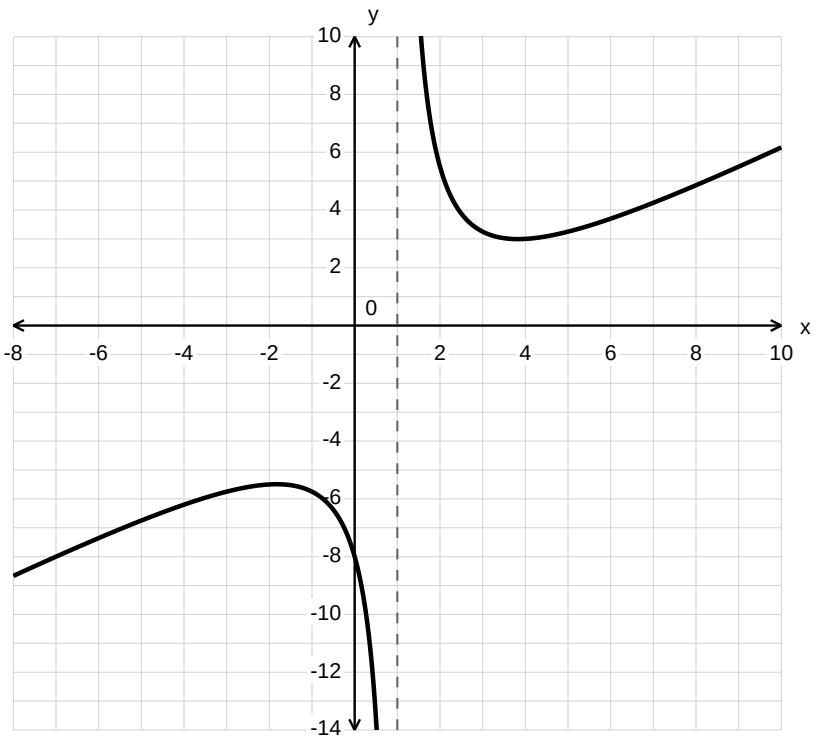
Your interpretation

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Problem 4. Connect the graph to degrees

The graph below represents a rational function *k*, written in reduced form.



Which relationship between the polynomial degrees must be true?

A. The denominator degree is one greater than the numerator degree.

B. The numerator and denominator have equal degrees.

C. The numerator degree is two greater than the denominator degree.

D. The numerator degree is one greater than the denominator degree.

Circle one answer. Explain how the graph supports your choice.

Work and reasoning

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Problem 5. Analyze and sketch

For the function below, factor and simplify. Keep the original restrictions.

$$g(x) = \frac{6x^2 + 3x - 18}{8x^2 - 28x + 24}$$

Factoring / simplified form: _____

_____

Domain restrictions: _____

x-intercept(s): _____

Vertical asymptote(s): _____

y-intercept: _____

Horizontal asymptote(s): _____

Hole coordinate(s): _____

Use the full-size coordinate grid on the next page for your sketch.

Additional factoring, calculations, and graph planning

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Problem 5 (continued). Your graph

Sketch the function from Problem 5. Label the intercepts, draw asymptotes as dashed lines, and mark any holes with open circles.

Each square represents 1 unit on both axes.

